

Math 109 – Precalculus

Section 6.1: Sinusoidal Graphs

From the unit circle to periodic functions

Last Time

We connected angles to points on a circle and used those points to evaluate trigonometric functions. In particular,

$$\cos(\theta) = \frac{x}{r}, \quad \sin(\theta) = \frac{y}{r}.$$

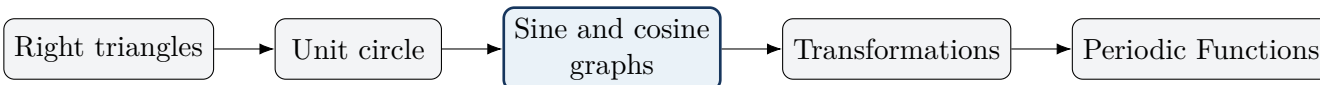
On the unit circle, the point at angle θ is

$$(\cos \theta, \sin \theta).$$

Today

By the end of today's lecture, you should be able to:

- explain how the unit circle produces the graphs of sine and cosine;
- identify the domain, range, amplitude, period, and midline of a sinusoidal function;
- describe how A , B , h , and k transform a sine or cosine graph;
- determine a possible equation from a sinusoidal graph.



1. Why Sinusoidal Graphs?

Think–Pair–Share

Imagine riding a Ferris wheel at a constant speed. Discuss how each quantity changes with time.

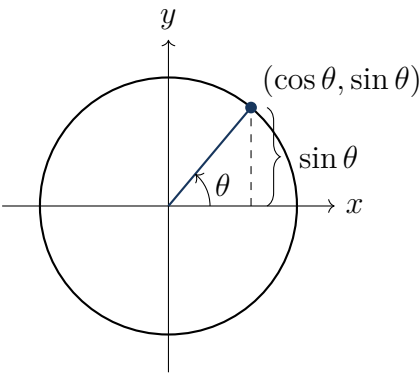
- your height above the ground;
- your distance from the center;
- your horizontal position;
- the angle through which the wheel has turned.

Which quantities repeat? Which remain constant? Which could be represented by a line?

The height of a rider rises and falls in a smooth, repeating pattern. Sine and cosine are natural functions for describing this kind of motion.

2. From the Unit Circle to the Sine Graph

The key idea is that the y -coordinate of a point on the unit circle becomes the height of the sine graph.

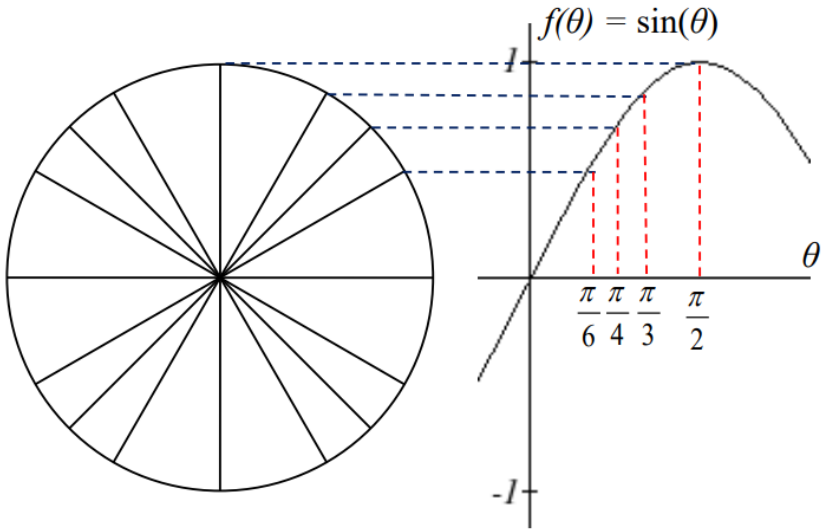


Strategy – Building the sine graph

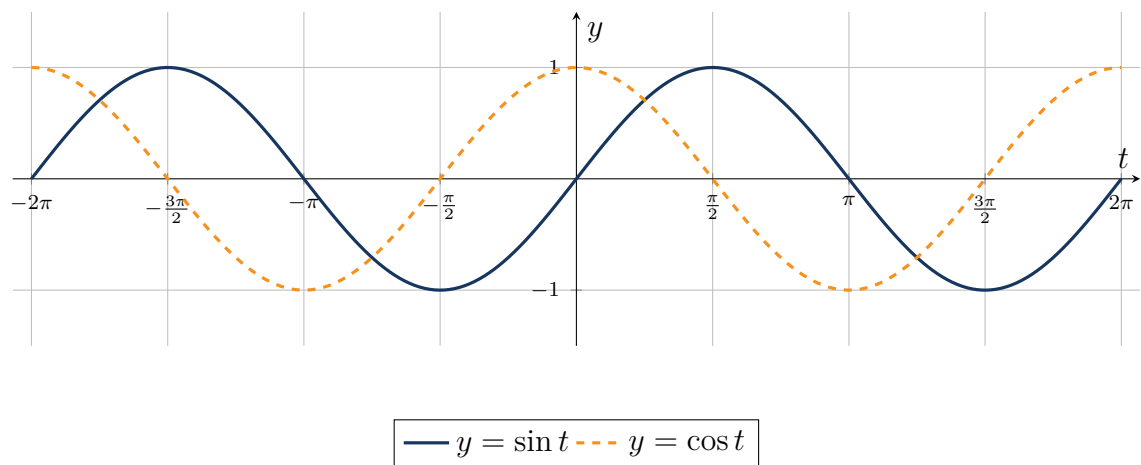
- For each angle t :
- 1. locate the corresponding point on the unit circle;
 - 2. record its y -coordinate, $\sin t$;
 - 3. plot the point $(t, \sin t)$ on a coordinate plane;
 - 4. connect the points with a smooth curve.

Complete the table using exact unit-circle values.

t	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
$\sin t$	—	—	—	—	—	—	—	—	—



3. The Basic Sine and Cosine Graphs



Characteristic	$y = \sin t$	$y = \cos t$
Domain	_____	_____
Range	_____	_____

Definition

A function is **periodic** when its output repeats after a fixed horizontal distance. In symbols, there is some $P > 0$ such that

$$f(t + P) = f(t)$$

for every allowable t . The smallest such positive value P is called the **period**.

Definition

The center value of a sinusoidal function (sin or cos) is the value that the function oscillates above and below. This center value is called the **midline** of the function.

The **amplitude** is the maximum vertical distance from the midline.

Characteristic	$y = \sin t$	$y = \cos t$
Period	_____	_____
Midline	_____	_____
Amplitude	_____	_____
Starting point at $t = 0$	_____	_____

Concept Check

How are the sine and cosine graphs related? Describe a horizontal shift that takes one graph to the other.

4. Transformations of Sine and Cosine

Use Desmos to compare each transformed function with $f(t) = \sin t$. Predict before graphing.

Transformed Function	Prediction	What actually changed?
$4 \sin t$		
$\sin t - 4$		
$\sin(5t)$		
$\sin(\frac{1}{5}t)$		

Think–Pair–Share

Is this the same as transformations of a line?

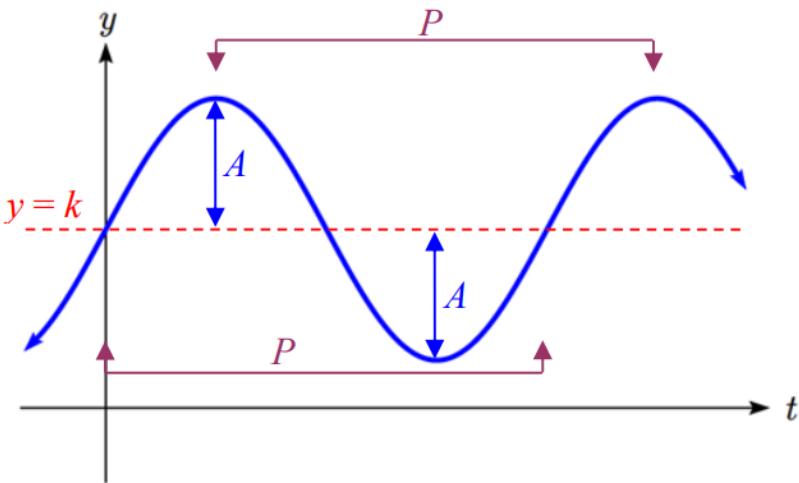
Definition – General transformed form

A transformed sine or cosine function can be written as

$$f(t) = A \sin(Bt) + k \quad \text{or} \quad f(t) = A \cos(Bt) + k.$$

The parameters have the following effects:

A	vertical stretch/compression; amplitude = $ A $
B	horizontal stretch/compression; related to the period $P = \frac{2\pi}{ B }$
k	vertical shift; midline $y = k$



Common Mistake

The coefficient B is *not* the period. It changes the period according to

$$P = \frac{2\pi}{|B|}.$$

A larger $|B|$ makes the graph repeat more quickly and therefore gives a shorter period.

Strategy – Reading a transformed equation

For $f(t) = A \sin(Bt) + k$ or $A \cos(Bt) + k$:

1. Find the amplitude: $|A|$.
2. Find the period: $2\pi/|B|$.
3. Find the midline: $y = k$.

Example 2: Read the characteristics from an equation

For each function, find the amplitude, period, and midline.

(a) $f(t) = 4 \sin\left(\frac{1}{3}t\right) - 2$

Amplitude = _____, Period = _____, Midline = _____.

(b) $g(t) = \frac{1}{2} \cos(\pi t) + 3$

Amplitude = _____, Period = _____, Midline = _____.

You Try: Read the characteristics

Find the amplitude, period, and midline of

$$h(t) = -3 \cos(4t) + 1.$$

5. Finding an Equation from a Graph

Strategy – From graph to formula

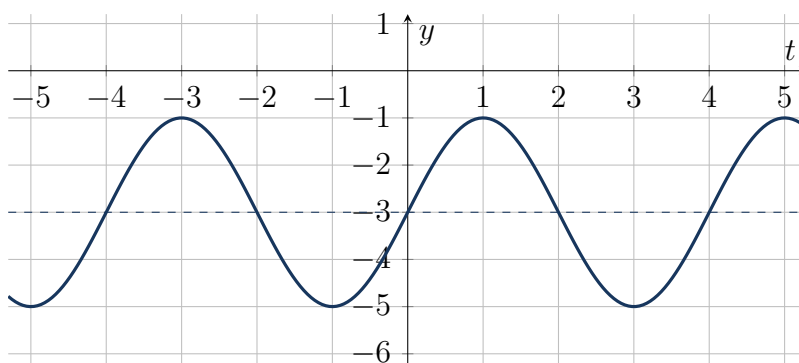
To build a sinusoidal model from a graph:

1. Identify the midline k .
2. Identify the amplitude A .
3. Measure one complete cycle to find the period P .
4. Compute $B = 2\pi/P$.
5. Write the formula.

Different correct formulas may describe the same graph.

Example 3: A graph with no horizontal shift needed

Find the amplitude, midline and period of the graph shown below. Use these to write a formula for the function



Midline: _____

Amplitude: _____

Period: _____

$$B = \frac{2\pi}{P} = \underline{\hspace{2cm}}$$

One possible equation is

$$f(t) = \underline{\hspace{4cm}}.$$

Think–Pair–Share

Could I have written the function shown above as a cosine function instead? What would I need to do?

6. Horizontal Shifts

Definition

Given an equation in the form

$$f(t) = A \sin(B(t - h)) + k \quad \text{or} \quad f(t) = A \cos(B(t - h)) + k,$$

the value h is the horizontal shift.

- $t - h$ shifts the graph **right** by h .
- $t + h = t - (-h)$ shifts the graph **left** by h .

Common Mistake

Always factor the coefficient of t before reading the horizontal shift. For example,

$$\sin\left(\frac{\pi}{4}t - \frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}(t - 1)\right),$$

so the shift is right by 1, not right by $\pi/4$.

Example 4: Find the Horizontal Shift

What is the horizontal shift in the function $f(t) = \sin(t + 8)$?

Example 5: Determine all transformations

Determine the amplitude, period, midline, and horizontal shift of

$$g(t) = 2 \sin(3t + 21) + 4.$$

Amplitude: _____

Period: _____

Midline: _____

Horizontal shift: _____.

You Try: Be careful when factoring

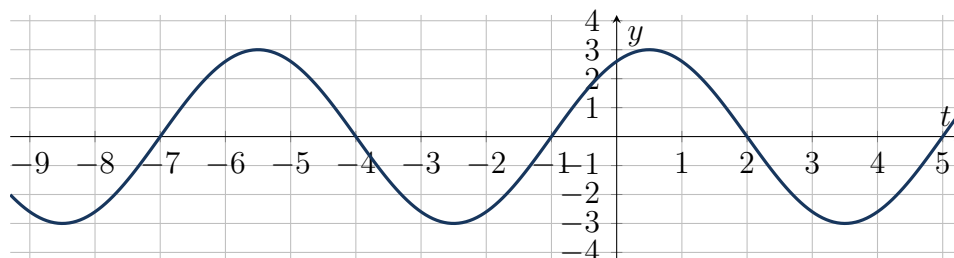
Determine the amplitude, period, midline, and horizontal shift of

$$p(t) = -5 \cos\left(2t - \frac{\pi}{3}\right) - 1.$$

7. All Together Now

Example 6: Find a possible formula from a graph

Find the formula for the function graphed below.



1. Midline: _____
2. Amplitude: _____
3. Period: _____, so $B =$ _____
4. One possible equation is

$$f(t) = \underline{\hspace{2cm}}.$$

Reflection

Why might two students produce different-looking equations for the same sinusoidal graph and both be correct?

Section 6.1 Summary

- The unit circle generates the sine and cosine graphs.
- Both basic functions have domain $(-\infty, \infty)$, range $[-1, 1]$, amplitude 1, midline $y = 0$, and period 2π .
- For $A \sin(B(t - h)) + k$ or $A \cos(B(t - h)) + k$:

$$\text{amplitude} = |A|, \quad P = \frac{2\pi}{|B|}, \quad \text{midline } y = k,$$

and h controls the horizontal shift.

- From a graph, use the maximum, minimum, and length of one cycle to determine the parameters.

Next Time

We will use the relationships between the trigonometric functions to understand the graphs of tangent, cotangent, secant, and cosecant.